

Mastering SOPs – Eight Golden Rules

**Patil Bhagyashri Sandip^{*1}, Patil Divyashree K², Sulbha G. Patil³, Bhushan R. Patil⁴,
Sunila.A.Patil⁵**

^{1,2,4} Student, ^{3,5} Assistant Professor

Department of (QA) P.S.G.V.P. Mandal's College of Pharmacy Shahada, Nandurbar

***Corresponding Author**

E-mail Id:-bhagyashripatil5853@gmail.com

ABSTRACT

Today, research data and information should be published online and, in theory, publicly accessible so that further research, products, or services can be developed. The long-term nature of research data is important for examining the economics of the production process. However, knowledge alone is not enough; measures to ensure reusability and reproducibility should cover the entire research lifecycle, from experimental design to data generation, quality control, statistical analysis, interpretation, and validation of results. Therefore, the quality of information, especially evidentiary information, is the main point that can be evidence for the customer. This information also increases the traceability and transparency of data and processes, increasing the reliability of results. Over the years, standards for data collection, analysis, and data have been developed with the leadership of researchers and organizations such as the Research Data Alliance (RDA). However, agreed-upon methods for routine research studies are still rare in the life sciences. Here, carefully designed documents such as standard operating procedures (SOPs) provide clear instructions and guidelines specifically designed to avoid inconsistencies that need to be renewed. They provide a workbook that explains step-by-step how to write an SOP as a starting point for proper research documentation.

Keywords:-Standard operating procedure, Steps.

INTRODUCTION

Today, technology is essential for the production and sharing of knowledge and scientific methods. Data availability is a key feature of an efficient, progressive, and ultimately self-improving research ecosystem that builds trust and becomes a key element of the integrity of research [1]. However, the results of successful integration can only be achieved if the data quality is reliable and reusable [2,3]. Despite this need, it has been found that overall, less than a third of (biomedical) data can be reproduced [4,5]. Further studies have shown that poor data management, unclear definitions, and reporting errors can inhibit reuse and reconstruction, thus affecting the use of

knowledge sharing and the reliability of findings [6–9]. Standard Operating Procedures (SOPs) for business processes have long existed to ensure efficiency, effectiveness, and consistency. SOPs ensure users follow procedures that comply with best practice standards. Additionally, the use of operational procedures ensures that procedures are continually reviewed and updated, allowing researchers in the same group or organization to replicate or reuse results again and again to extend the study or apply it to other studies. Despite their importance, only in recent years has the need to use models in life sciences become important in terms of the quality and reproducibility of research results [9]. Data

quality and reproducibility issues create new scientific and social challenges for researchers, universities, research institutes, infrastructure, capital resources, and wider society [10]. Around 2016, the scientific community began to discuss the reproducibility issue, its serious impact on business, and the reliability of scientific technology [9,11]. An index is proposed to support the scientific community in solving problems related to the need for new tools and techniques to improve the compatibility of the measurement model [13–14] and use models in modern science to improve research and data reuse. Since then, many advances have been made in various areas of biological research [15–20].

There are many suggestions that provide guidelines for managing returns using simple rules. These suggestions include the use of error messages on generated data, which can be used to evaluate the suitability and reliability of a profile in relation to the generated results. Additionally, links should be used when publishing research results. Links make it easier to track results and repeat successful experiments and can be linked to the classification of errors above. Information about the input data and the software versions used fits the connections perfectly and plays an important role in generating the results. Finally, adopting these simple rules in regular research in SOP mode leads to a clear view of the results and the decision to accept the research[21].

Essential Components of an SOP [22]

Since SOPs cover a wide range of industries, workplaces, and processes, standards may vary. But there are a few important factors that allow you to get the most out of it. A well-written SOP should include the following important points:

Title: This may seem obvious, but the Title is clear and clarifies the purpose of the Process, it is an important part of organizing the content for the reader.

Purpose: To provide an overview – a brief description explaining the purpose and function of the project. Responsibilities of Volunteers. A brief description of the task, including precautions, warnings, or important steps. Any supporting information, guidance or standards.

Definitions: Clarify any technical terms, acronyms, or abbreviations used in the documentation.

Responsibilities: Identify the roles and responsibilities of the personnel involved in performing the procedure.

Materials and Equipment: List all necessary materials, tools, software, and equipment needed to perform the procedure.

Procedure: The core of any SOP – offers a comprehensive explanation of the steps needed to finish the process, including any important steps, warnings, or precautions.

Documentation: Include instructions on how to record data, maintain logs, or fill out forms related to the procedure.

References: Give any supporting archives, rules, or administrative guidelines connected with the technique.

Approval and Revision History: Show who approved the process, the date it started, and any changes made to the document.

How to write an SOP: Step by step guide

For pioneers who need to advance the proficiency, consistency and nature of their groups' work, it is fundamental to make a standard method. Here is a one-size-fits-all manual for making a SOP to assist you with considering every contingency.

Step 1: Identify the Process

If you're here, chances are you already have a specific process within your organization that requires standardization. This could be a task causing operational delays, inconsistency in execution, or a procedure vital for compliance and safety. Pinpointing the correct process ensures

that your SOP can make the most significant impact.

Step 2: Assemble a Team

Engage individuals who directly perform, oversee, and rely on the outcomes of the process. Their firsthand knowledge and insights are invaluable in crafting documentation that is both accurate and practical. Embracing diverse perspectives can also uncover areas for enhancement that may have been overlooked.

Step 3: Establish Scope and Objectives

Clearly define what the SOP will encompass and what it aims to accomplish. This clarity helps prevent scope creep and maintains the document's focus and efficiency. Whether the goal is to enhance safety, ensure quality, or streamline a process, having well-defined objectives is essential.

Step 4: Deconstruct the Process

Detail each step of the process comprehensively. Begin with a broad overview before delving into specific actions. Use straightforward language to ensure comprehension across all organizational levels. Additionally, include any necessary materials, tools, or software required to execute each step.

Step 5: Delegate Roles and Responsibilities

Define who is responsible for each step of the process in a clear way. This straightforwardness destroys disarray and lays out clear lines of liability. It's not just about task appointment; It's about setting clear goals and giving team members authority.

Step 6: Review and Trial

Prior to concluding the SOP, going through an exhaustive survey and testing stage with the team is fundamental. This time for testing can uncover useful deterrents or regions needing upgrade. Criticism from day to day clients is basic for creating a report that is easy to use and compelling.

Step 7: Implementation and Training

Following SOP revisions, educate your team through training sessions, demonstrations, or written guidelines. This ensures that everyone comprehends and follows the SOP accurately. It's crucial that all involved individuals understand and feel confident with the procedures outlined. This step is vital for fostering acceptance and adherence.

Step 8: Monitoring and Revisions

An SOP should adapt alongside your evolving processes. Establish regular evaluations to gauge its efficacy and implement necessary revisions. This continuous improvement cycle ensures the SOP remains relevant and beneficial as your business grows and changes.

Benefits of SOPs[23]

1. Ensuring Consistency and Reliability in Product and Service Delivery
2. Enhancing Quality Assurance and Safety Standards
3. Saving Time and Cost
4. Streamlining Employee Management Processes
5. Preventing Knowledge Erosion
6. Facilitating Audits and Compliance
7. Promoting Brand Standardization
8. Optimizing Customer Service
9. Facilitating Task Transferability
10. Meeting Legal Obligations and Fostering Communication and Operational Consistency.

CONCLUSION

Nowadays, advanced technology and computational methods have become an important part of daily experimental studies. To maintain the highest quality data generated and avoid reproducibility issues, researchers should use FAIR data, appropriate DMPs, and appropriate data. The absence of reproducibility of lab concentrates on upsets the outcome of logical examination that can be broadly utilized by mainstream researchers. One way to improve this is to ensure consistency and traceability between

existing models and laboratory tests; This can be achieved through well-written and clear working procedures.

REFERENCES

1. Hardwicke TE, Mathur MB, MacDonald K, Nilsson G., Banks GC, Kidwell MC, et al. Data availability, reusability, and analytic reproducibility: evaluating the impact of a mandatory open data policy at the journal *Cognition*. *R Soc Open Sci*. 2018. August 15;5(8):180448 10.1098/rsos.180448 ; PubMed PMCID: PMC6124055 [PMC free article] [PubMed] [CrossRef] [Google Scholar]
2. Benefits of sharing. Editorial. *Nature*. 2016. February 11;530(7589):129 10.1038/530129a [PubMed] [CrossRef] [Google Scholar]
3. Doucet M, Becker KF, Björkman J, Bonnet J, Clément B, Daidone M-G, et al. Quality Matters: Annual Conference of the National Infrastructures for Biobanking. *Biopreserv Biobank* 2017. June;15(3):270–276. 10.1089/bio.2016.0053 ; PubMed PMCID: PMC5586151 [PMC free article] [PubMed] [CrossRef] [Google Scholar]
4. Borgman CL. Big data, little data, no data: Scholarship in the networked world. The MIT Press; 2015. [Google Scholar]
5. Harris JK, Wondmeneh SB, Zhao Y, Leider JP. Examining the Reproducibility of 6 Published Studies in Public Health Services and Systems Research. *J Public Health Manag Pract*. 2019. 10.1097/PHH.0000000000000694 [PubMed] [CrossRef] [Google Scholar]
6. Baker M. Irreproducible biology research costs put at \$28 billion per year. *Nature News* 2015. June:9. Doi.org/10.1038/nature.2015.17711 [Google Scholar]
7. Baker M. 1,500 scientists lift the lid on reproducibility. *Nature*. 2016. May 6;533(7604):452–4. 10.1038/533452a [PubMed] [CrossRef] [Google Scholar]
8. Freedman LP, Cockburn IM, Simcoe TS. The Economics of Reproducibility in Preclinical Research. *PLoS Biol*. 2015. 10.1371/journal.pbio.1002165 ; Correction: The Economics of Reproducibility in Preclinical Research. *PLoS Biol*. 2018. PMID: 29634726 [PMC free article] [PubMed] [CrossRef] [Google Scholar]
9. Allison D, Brown A, George B, Kaiser KA. Reproducibility: A tragedy of errors. *Nature* 2016. February 4;530(7588):27–9. 10.1038/530027a ; PubMed PMCID: PMC4831566 [PMC free article] [PubMed] [CrossRef] [Google Scholar]
10. Baker M. How quality control could save your science. *Nature*. 2016. January 28;529(7587):456–8. 10.1038/529456a [PubMed] [CrossRef] [Google Scholar]
11. European Commission. Directorate-General for Research and Innovation (European Commission), PwC EU Services. Cost-Benefit analysis for FAIR research data—Cost of not having FAIR research data. 2019. Doi.org/10.2777/706548
12. Research Data Alliance (RDA) [cited 2020 Feb 6]. Available from: <https://www.rd-alliance.org/>
13. Wilkinson M, Dumontier M, Aalbersberg I, Appleton G, Axton M, Baak A, et al. The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data*. 2016. March 15;3:160018 10.1038/sdata.2016.18 [PMC free article] [PubMed] [CrossRef] [Google Scholar]

14. cost-charme.eu [Internet]. COST Action Harmonising standardisation strategies to increase efficiency and competitiveness of European life-science research (CHARME); c2016-2020 [cited 2020 Apr 7]. Available from: <https://www.cost-charme.eu/>. Available from: <https://www.cost-charme.eu/>
15. Bergmann FT, Hoops S, Klahn B, Kummer U, Mendes P, Pahle J, et al. COPASI and its applications in biotechnology. *J Biotechnol.* 2017. November 10;261:215–220. 10.1016/j.jbiotec.2017.06.1200 [PMC free article] [PubMed] [CrossRef] [Google Scholar]
16. von Kamp A, Thiele S, Hädicke O, Klamt S. Use of CellNetAnalyzer in biotechnology and metabolic engineering. *J Biotechnol.* 2017. November 10;261:221–228. 10.1016/j.jbiotec.2017.05.001 [PubMed] [CrossRef] [Google Scholar]
17. Wittig U, Rey M, Weidemann A, Kania R, Müller W. SABIO-RK: an updated resource for manually curated biochemical reaction kinetics. *Nucleic Acids Res.* 2018. January 4;46(D1):D656–D660. 10.1093/nar/gkx1065 [PMC free article] [PubMed] [CrossRef] [Google Scholar]
18. Wolstencroft K, Owen S, Krebs O, Nguyen Q, Stanford NJ, Golebiewski M, et al. SEEK: a systems biology data and model management platform. *BMC Syst Biol.* 2015. July 11;9:33 10.1186/s12918-015-0174-y [PMC free article] [PubMed] [CrossRef] [Google Scholar]
19. Rocca-Serra P, Sansone SA. Experiment design driven FAIRification of omics data matrices, an exemplar. *Sci Data.* 2019. December 12;6(1):271 10.1038/s41597-019-0286-0 ; PubMed PMID: PMC6908569 [PMC free article] [PubMed] [CrossRef] [Google Scholar]
20. go-fair.org [Internet]. GO FAIR; [cited 2020 Apr 7]. Available from: <https://www.go-fair.org/>
21. Conroy G. Q&A: 5 simple ways to make your research more reproducible. Small details can make a big difference. *Nature Index.* 31 October 2019; [cited 2020 Apr 7]. Available from: <https://www.natureindex.com/news-blog/david-sholl-five-simple-ways-to-make-your-research-more-reproducible> [Google Scholar]
22. <https://www.getguru.com/reference/what-is-an-sop-definition-examples-how-to-make-one>
23. <https://scribeshow.com/library/benefits-of-sops>